22 There are only r red counters and g green counters in a bag.

A counter is taken at random from the bag.

The probability that the counter is green is $\frac{3}{7}$

The counter is put back in the bag.

2 more red counters and 3 more green counters are put in the bag.
A counter is taken at random from the bag.

The probability that the counter is green is $\frac{6}{13}$

Find the number of red counters and the number of green counters that were in the bag originally.

Initially green : red = 3 : 4.

1 mark
Use 3/7 to link green and red.

Let green = 3x and red = 4x.

After adding: green = 3x + 3,

red = 4x + 2; total = 7x + 5.

1 mark
Form the new probability.

(3x + 3)/(7x + 5) = 6/13

13(3x + 3) = 6(7x + 5)

1 mark
Clear the denominators.

39x + 39 = 42x + 30; x = 3.

1 mark
Solve for x.

Red = 4 × 3 = 12; green = 3 × 3 = 9.

1 mark
12 red and 9 green originally.

red counters 12

green counters 9

(Total for Question 22 is 5 marks)